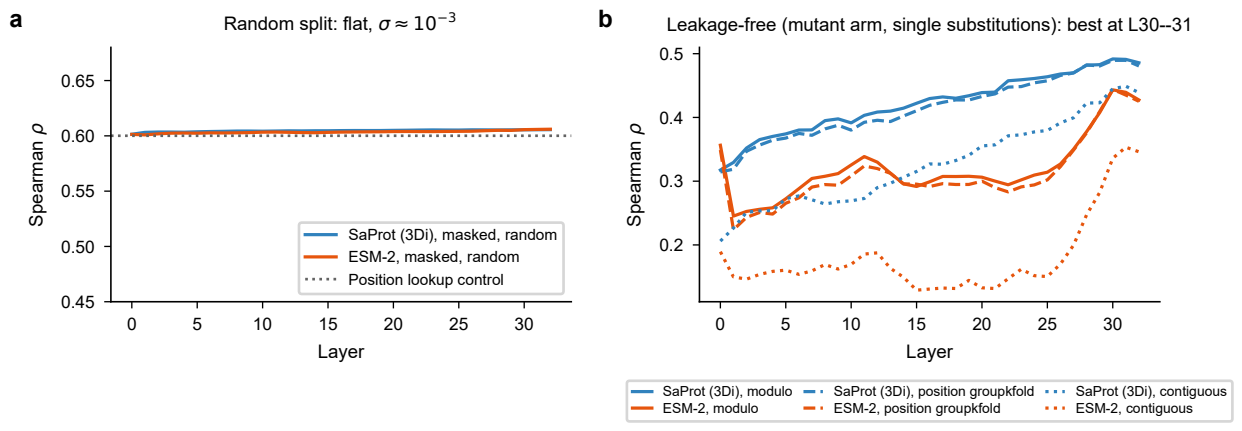

a
Random split: flat, $\sigma \approx 10^{-3}$
Spearman ρ
0.65
0.60
0.55
0.50
0.45
0
5
10
15
20
25
30
Layer
SaProt (3Di), masked, random
ESM-2, masked, random
Position lookup control
b
Leakage-free (mutant arm, single substitutions): best at L30--31
Spearman ρ
0.5
0.4
0.3
0.2
0
5
10
15
20
25
30
Layer
SaProt (3Di), modulo
ESM-2, modulo
SaProt (3Di), position groupkfold
ESM-2, position groupkfold
SaProt (3Di), contiguous
ESM-2, contiguous